LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Unit step response of a control system, Transient response, Steady state response, Max overshoot, Rise time, Delay time and settling time.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Unit step response of a control system, Transient response, Steady state response, Max overshoot, Rise time, Delay time and settling time.
5. Specific Objectives	20.1.1.1 Trainee will be able to classify control systems, feedback and its effect.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Analysis of step response plots to identify transient and steady state parameters; Safety: No physical equipment handling; Supervision: Instructor monitors plot interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define a control system.
(b) Explain the function of feedback in a control loop.
(c) Differentiate between open-loop and closed-loop systems.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Unit step input definition • Transient response characteristics • Steady state response definition	L & Ds	PPT, IB	15 min
(b)	• Maximum overshoot calculation • Rise time measurement • Delay time and settling time	L & Ds	PPT, PDF	20 min
(c)	• Effect of damping ratio on response • Impact of feedback on stability • Summary of time domain specs	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) How does increasing damping ratio affect maximum overshoot?
(b) Define settling time and explain its significance in system performance.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of rise time and settling time apply to the stabilization of an aircraft autopilot system, linking back to the introduction of ATE in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to ATE and its uses.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min